Multiple Linear Regression Case Study : Predicting used car selling price

Background

This dataset contains information about used cars. Each row in the dataset represents a unique listing of a used car, and there are several features that describe different aspects of these cars.

Objective:

The objective is to develop a predictive model which can be used to estimate the selling price of used cars based on the various features.

Import Libraries

```
library(dplyr)
library(reshape2)
library(caret)
library(car)
library(nortest)
```

Load Data

```
data<-read.csv("C:/Users/SANKHYA/Downloads/Vehicle Data.csv",header=T)
head(data)
```

```
                           name year selling_price km_driven   fuel seller_type
1        Maruti Swift Dzire VDI 2014        450000    145500 Diesel  Individual
2  Skoda Rapid 1.5 TDI Ambition 2014        370000    120000 Diesel  Individual
3      Honda City 2017-2020 EXi 2006        158000    140000 Petrol  Individual
4     Hyundai i20 Sportz Diesel 2010        225000    127000 Diesel  Individual
5        Maruti Swift VXI BSIII 2007        130000    120000 Petrol  Individual
6 Hyundai Xcent 1.2 VTVT E Plus 2017        440000     45000 Petrol  Individual
  transmission        owner mileage engine max_power seats
1       Manual  First Owner   23.40   1248     74.00     5
2       Manual Second Owner   21.14   1498    103.52     5
3       Manual  Third Owner   17.70   1497     78.00     5
4       Manual  First Owner   23.00   1396     90.00     5
5       Manual  First Owner   16.10   1298     88.20     5
6       Manual  First Owner   20.14   1197     81.86     5
```

Data Description :

• name: The car’s make and model.

• year: The year in which the car was manufactured.

• selling_price: The price at which the car was sold.

• km_driven: The number of kilometers the car has been driven.

• seller_type: The type of seller (e.g., Individual, Dealer,TrustmarkDealer).

• transmission: The type of transmission (e.g., Manual, Automatic).

• owner: The history of car ownership (e.g., First Owner, Second Owner,TestDriveCar,ThirdOwner,Fourth&AboveOwner).

• mileage: The car’s mileage in kilometers per liter (kmpl).

• engine: The engine capacity in cubic centimeters (cc).

• max_power: The maximum engine power in bhp (brake horsepower).

• seats: The number of seats in the car.

Summarizing Data And Checking Missing Values

```
summary(data)
```

```
     name               year      selling_price        km_driven
 Length:8128        Min.   :1983   Min.   :   29999   Min.   :      1
 Class :character   1st Qu.:2011   1st Qu.:  254999   1st Qu.:  35000
 Mode  :character   Median :2015   Median :  450000   Median :  60000
                    Mean   :2014   Mean   :  638272   Mean   :  69820
                    3rd Qu.:2017   3rd Qu.:  675000   3rd Qu.:  98000
                    Max.   :2020   Max.   :10000000   Max.   :2360457

     fuel           seller_type        transmission          owner
 Length:8128        Length:8128        Length:8128        Length:8128
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character




    mileage          engine       max_power          seats
 Min.   : 0.00   Min.   : 624   Min.   :  0.00   Min.   : 2.000
 1st Qu.:16.78   1st Qu.:1197   1st Qu.: 68.05   1st Qu.: 5.000
 Median :19.30   Median :1248   Median : 82.00   Median : 5.000
 Mean   :19.42   Mean   :1459   Mean   : 91.52   Mean   : 5.417
 3rd Qu.:22.32   3rd Qu.:1582   3rd Qu.:102.00   3rd Qu.: 5.000
 Max.   :42.00   Max.   :3604   Max.   :400.00   Max.   :14.000
 NA's   :221     NA's   :221    NA's   :216      NA's   :221
```

```
colSums(is.na(data))
```

```
         name          year selling_price     km_driven          fuel
            0             0             0             0             0
  seller_type  transmission         owner       mileage        engine
            0             0             0           221           221
    max_power         seats
          216           221
```

```
#Drop NAs
data<-na.omit(data)
```

Correlation Analysis and Heatmap

```
CM_data<- data %>% select(selling_price,year,km_driven,mileage,engine,max_power,seats)
correlation_matrix<-round(cor(CM_data),2)
melted_CM<-melt(correlation_matrix)

ggplot(melted_CM, aes(x=Var1,y=Var2,fill=value))+geom_tile()+
  geom_text(aes(Var1, Var2, label=value),color="black",size=4)+
  scale_fill_gradient2(low="red", mid="white", high="blue")+labs(title = "",x="",y="")
```

Observations:

• Selling_price has highest correlation with max_power(strong +ve) followed by engine size (+ve).

Linear Regression Analysis

```
set.seed(123)
# Excluding name and year from analysis
#Split data into training and test sets (e.g., 80% training, 20% test)
index<-createDataPartition(data$selling_price, p=0.8, list=FALSE)

traindata<-data[index,]
testdata<-data[-index,]

#Define your formula
data_model<- lm(selling_price~km_driven+seller_type+transmission+owner+mileage+max_power+
                  engine+seats,data=traindata)
data_model
```

```
Call:
lm(formula = selling_price ~ km_driven + seller_type + transmission + 
    owner + mileage + max_power + engine + seats, data = traindata)

Coefficients:
                (Intercept)                    km_driven  
                 -5.402e+05                   -2.059e+00  
      seller_typeIndividual  seller_typeTrustmark Dealer  
                 -2.600e+05                   -3.494e+05  
         transmissionManual    ownerFourth & Above Owner  
                 -4.715e+05                   -1.095e+05  
          ownerSecond Owner          ownerTest Drive Car  
                 -9.270e+04                    2.267e+06  
           ownerThird Owner                      mileage  
                 -1.059e+05                    2.945e+04  
                  max_power                       engine  
                  1.343e+04                    1.459e+02  
                      seats                               
                 -2.051e+03                               
```

```
#Print the model summary
summary(data_model)
```

```
Call:
lm(formula = selling_price ~ km_driven + seller_type + transmission + 
    owner + mileage + max_power + engine + seats, data = traindata)

Residuals:
     Min       1Q   Median       3Q      Max 
-2551262  -198382    15387   161534  3974857 

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 -5.402e+05  7.050e+04  -7.662 2.11e-14 ***
km_driven                   -2.059e+00  1.301e-01 -15.822  < 2e-16 ***
seller_typeIndividual       -2.600e+05  1.878e+04 -13.841  < 2e-16 ***
seller_typeTrustmark Dealer -3.494e+05  3.907e+04  -8.943  < 2e-16 ***
transmissionManual          -4.715e+05  2.237e+04 -21.079  < 2e-16 ***
ownerFourth & Above Owner   -1.095e+05  4.154e+04  -2.636  0.00841 ** 
ownerSecond Owner           -9.270e+04  1.448e+04  -6.401 1.66e-10 ***
ownerTest Drive Car          2.267e+06  2.681e+05   8.455  < 2e-16 ***
ownerThird Owner            -1.059e+05  2.527e+04  -4.192 2.81e-05 ***
mileage                      2.945e+04  1.850e+03  15.914  < 2e-16 ***
max_power                    1.343e+04  2.863e+02  46.907  < 2e-16 ***
engine                       1.459e+02  2.392e+01   6.102 1.11e-09 ***
seats                       -2.051e+03  8.675e+03  -0.236  0.81310    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 463000 on 6314 degrees of freedom
Multiple R-squared:  0.6793,	Adjusted R-squared:  0.6787 
F-statistic:  1114 on 12 and 6314 DF,  p-value: < 2.2e-16
```

Observations:

• In the LM Regression Results, the model's R-squared value is 0.67, indicating that approximately 67% of the variability in selling_price is explained by the model.

• The coefficients for various predictor variables suggest their respective impact on the selling price. Notably, the variables "mileage" , "max_power" and "engine" have substantial positive effects, while "km_driven" has a negative effect.

Variance Inflation Factor (VIF) Analysis (checking Multicollinearity)

```
vif(data_model)
```

```
                 GVIF Df GVIF^(1/(2*Df))
km_driven    1.322788  1        1.150125
seller_type  1.283487  2        1.064383
transmission 1.669249  1        1.291994
owner        1.234548  4        1.026688
mileage      1.636023  1        1.279071
max_power    3.104998  1        1.762100
engine       4.266799  1        2.065623
seats        2.056969  1        1.434214
```

Observations:

- The multicollinearity problem is resolved as all VIF's are less than 5.

Model Refinement: Excluding Insignificant Variable (seats)

```
#Revised Formula -- Excluding max power
data_model1<- lm(selling_price~km_driven+seller_type+transmission+owner+mileage+engine+max
_power, data=traindata)
data_model1
```

```


Call:
lm(formula = selling_price ~ km_driven + seller_type + transmission +
    owner + mileage + engine + max_power, data = traindata)

Coefficients:
                (Intercept)                    km_driven
                 -5.493e+05                   -2.059e+00
      seller_typeIndividual  seller_typeTrustmark Dealer
                 -2.602e+05                   -3.490e+05
         transmissionManual    ownerFourth & Above Owner
                 -4.723e+05                   -1.092e+05
          ownerSecond Owner          ownerTest Drive Car
                 -9.238e+04                    2.267e+06
           ownerThird Owner                      mileage
                 -1.054e+05                    2.952e+04
                     engine                     max_power
                  1.427e+02                    1.345e+04
```

```
#Print the model summary
summary(data_model1)
```

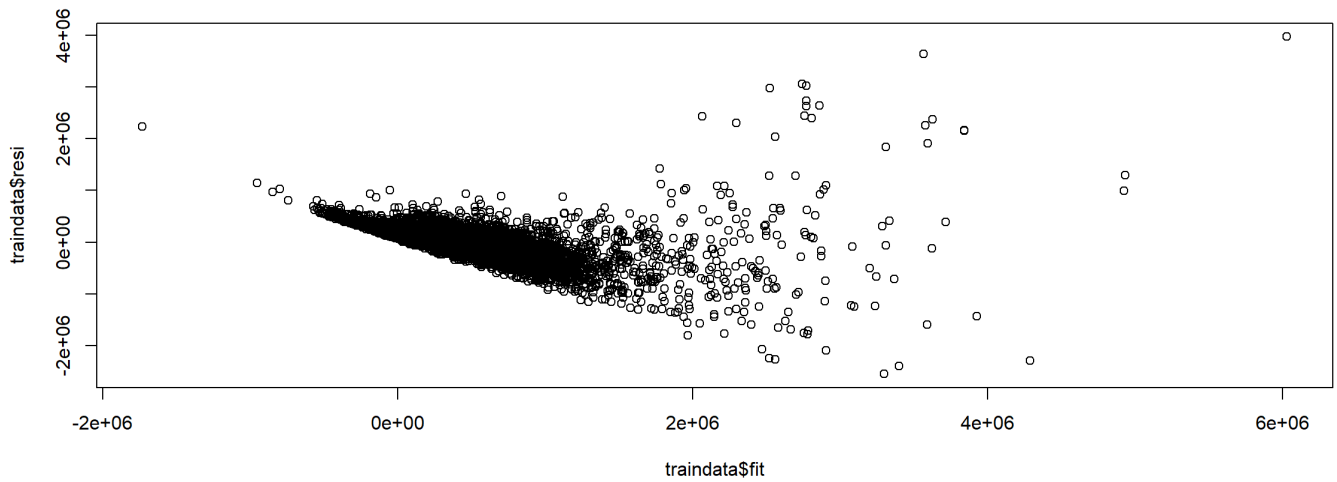
```
Call:
lm(formula = selling_price ~ km_driven + seller_type + transmission + 
    owner + mileage + engine + max_power, data = traindata)

Residuals:
     Min       1Q   Median       3Q      Max 
-2549323  -198463    15844   161640  3970748 

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 -5.493e+05  5.903e+04  -9.305  < 2e-16 ***
km_driven                   -2.059e+00  1.301e-01 -15.824  < 2e-16 ***
seller_typeIndividual       -2.602e+05  1.875e+04 -13.874  < 2e-16 ***
seller_typeTrustmark Dealer -3.490e+05  3.903e+04  -8.942  < 2e-16 ***
transmissionManual          -4.723e+05  2.215e+04 -21.325  < 2e-16 ***
ownerFourth & Above Owner   -1.092e+05  4.151e+04  -2.630  0.00857 ** 
ownerSecond Owner           -9.238e+04  1.442e+04  -6.407 1.59e-10 ***
ownerTest Drive Car          2.267e+06  2.681e+05   8.455  < 2e-16 ***
ownerThird Owner            -1.054e+05  2.518e+04  -4.187 2.87e-05 ***
mileage                      2.952e+04  1.820e+03  16.217  < 2e-16 ***
engine                       1.427e+02  1.962e+01   7.273 3.93e-13 ***
max_power                    1.345e+04  2.723e+02  49.384  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 463000 on 6315 degrees of freedom
Multiple R-squared:  0.6793,	Adjusted R-squared:  0.6787 
F-statistic:  1216 on 11 and 6315 DF,  p-value: < 2.2e-16
```

Observations:

• After excluding “seats” from the model , the R-squared value is still the same, indicating that approximately 67% of the variability in selling_price is explained by the revised model.

Residual Analysis and Scatter Plot

```
#Calculate and store the residuals
traindata$resi<-residuals(data_model1)
#Assign the fitted values
traindata$fit<-fitted(data_model1)
#Create a scatter plot of residuals vs. predicted values
plot(traindata$fit,traindata$resi)
```

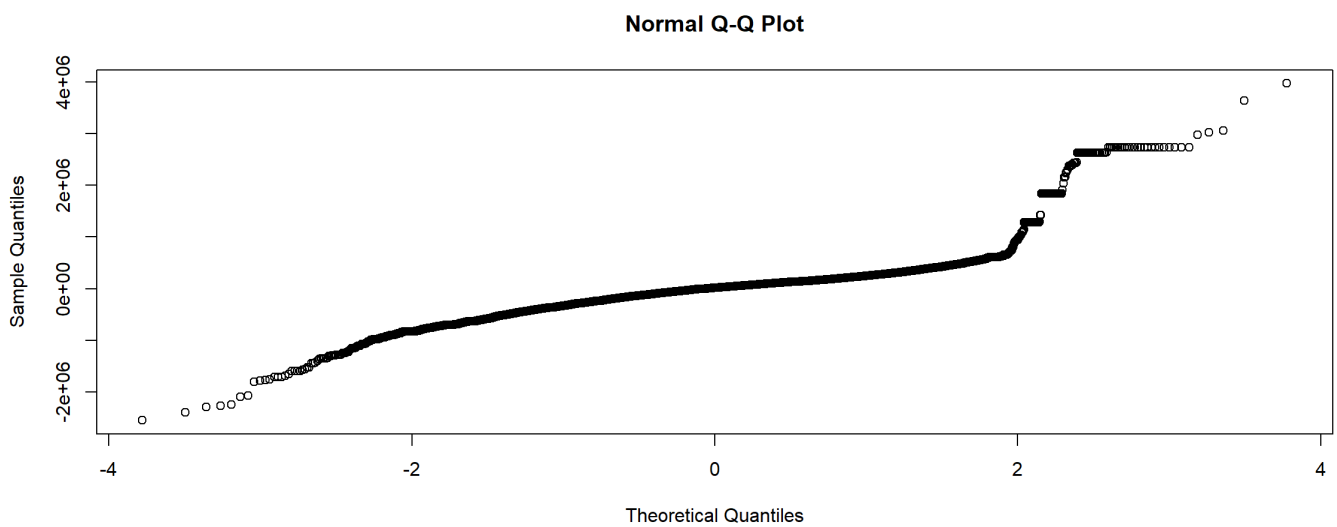


Observations:

It is observed that residuals are randomly distributed and uncorelated with predicted values.

Checking Normality

```
qqnorm(traindata$resi)
```




```
#n<-length(traindata$resi)
#Randomly sample 5000 observations from data
subset_data<-sample(traindata$resi, 5000)
shapiro.test(subset_data)
```

```
    Shapiro-Wilk normality test

data:  subset_data
W = 0.79226, p-value < 2.2e-16
```

```
lillie.test(traindata$resi)
```

```
    Lilliefors (Kolmogorov-Smirnov) normality test

data:  traindata$resi
D = 0.1423, p-value < 2.2e-16
```

Observations :

Although normality of errors is not established we will proceed to evaluate the model performance.

Model Validation: Holdout Method using RMSE

```
traindata$resi<-residuals(data_model1)
RMSE<-sqrt(mean(traindata$resi**2))
RMSE
```

```
[1] 462557.6
```

RMSE is 462557.6

```
testdata$pred<-predict(data_model1,testdata)
testdata$resi<-(testdata$selling_price-testdata$pred)
RMSEtest<-sqrt(mean(testdata$resi**2))
RMSEtest
```

```
[1] 481352.7
```

RMSE is 481352.7

Observations :

The similarity in RMSE between the test and train data suggests that the regression model generalizes reasonably well, with consistent prediction accuracy on both datasets. This implies

that the model is stable and not overfitting to the training data.

NOTE :

While we possess data dating back to 1981, it is important to recognize that the concept of ‘value for money’ has evolved over time. To account for this evolution, we can calculate the present selling price by considering the historical selling price and the number of years since the item was originally priced. This calculation will involve an annual increase of 5% in the selling price to adjust for inflation and changing market conditions

Formula :

Calculate n = 2022-year

Present selling price = historical selling price * ((1+0.5)^n)